Zebra CV Developer II — Recruiter Screen Guide

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Role snapshot

This is a mid-level C++ engineering role: take Zebra's computer-vision prototypes and make them fast, tested product code on hardware from PCs down to smart cameras. It is one or two levels below your experience, which shapes the salary and "why this role" answers below.

Item	Detail
Title	Computer Vision Software Developer, Professional II
Req ID	JR103160 (posted ~29 Sep 2026)
Location	Montreal, hybrid: minimum 3 days/week in office. Almost certainly the former Matrox Imaging site in Dorval (confirm on the call)
Pay	$90,960 – $136,440 base (assumed CAD) + 10% target annual cash bonus
Minimum bar	BSc/MSc CE/CS, 2+ yrs, strong C/C++, basic CV in 2+ of: detection/tracking/recognition, multi-view geometry, OCR, calibration, 3D, SfM/SLAM, activity recognition
Preferred	Reusable large-scale CV algorithms, parallel/vectorized compute, applied maths, Python prototyping, French, CUDA/OpenCL, GenAI dev tools, VLMs/ViTs/CNNs
Core work	Prototype → production C++; per-platform optimization (x86, GPU, multi-core ARM, DSP / AI accelerator) under performance, power and size budgets; code review; automated tests; architecture docs

The English and French versions differ slightly: the French adds deep-learning prototypes and lists "AI accelerator" where the English says DSP. Read that as on-device neural-network inference being part of the job.

Company check

Verdict: the posting is legitimate and the company is financially healthy. Machine vision is a stated growth priority, but the unit went through restructuring in early 2026, so team stability is worth one polite question.

Area	Finding	What it means for you
Business	Zebra Technologies (NASDAQ: ZBRA), Lincolnshire, IL. Barcode printers and scanners, rugged mobile computers, RFID, machine vision. CEO Bill Burns	Large, stable hardware + software company; CV is one product line among several
Financials	Q2 2026 (quarter ended 4 Jul 2026): sales $1.557B, up 20.4% (9.2% organic); net income $233M; full-year outlook raised (8-K). Part of the margin jump came from $73M in tariff recoveries	No financial distress. Q3 results are due around late October
Montreal site	Zebra bought Matrox Imaging (Dorval, founded 1976) in 2022 for US$875M; Matrox Imaging had ~US$100M revenue (Wiley Industry News)	This team is the old Matrox Imaging group: decades of industrial machine-vision software, not a startup-style AI lab
Product you'd touch	Aurora Imaging Library (formerly Matrox Imaging Library): C++/C#/Python SDK for 2D and 3D image analysis, plus deep-learning add-ons for unsupervised anomaly detection, OCR and object detection on NVIDIA/Intel GPUs (BusinessWire)	Maps directly onto your care-label QC and OCR work. Learn the product names before the call
3D vision	Acquired Photoneo (3D sensors, vision-guided robotics) in Feb 2025 for $62M (10-Q)	Explains "3D data processing" and "multiple view geometry" in the JD

Area	Finding	What it means for you
Restructuring	Exited robotics; sold it to Skild AI in Q1 2026. A "2025 Productivity Plan" reorganization was substantially completed in Q2 2026 (10-Q)	Cuts are mostly behind them per filings; machine vision was named a priority area
Layoffs	WARN filing: 75 jobs in San Jose, Feb 2026 (WARNTracker). An anonymous post claims Machine Vision was "hit hard" in Feb 2026 (TheLayoff) — unverified	Hiring at two levels now (a CV Developer I is also open in Montreal) suggests rebuilding, but ask about team size
Machine vision outlook	Management guided MV back to growth in 2026; Q2 manufacturing demand was strong in electronics and pharma (Q4 2025 call)	Positive signal for the unit's budget
Culture	Glassdoor 3.7/5, 67% recommend, 77% CEO approval. A Feb 2026 review praises the Montreal office but cites 4–6 h/day of meetings, layered approvals and direction changes (Glassdoor)	Typical large-company friction; probe decision speed in later rounds, not this call
Legitimacy	Workday ATS, matching req ID, same role listed on Glassdoor. Zebra states recruiters only use @zebra.com addresses and never collect personal data outside the ATS	Check the recruiter's email domain. Give SIN or banking details to nobody at this stage

Fit map

You clear every minimum requirement; the core of the job (prototype → optimized C++ on constrained hardware) is exactly what you did at MAS and in the postdoc. The soft spots are GPU kernels, SIMD, VLMs/ViTs and French.

JD asks for	Your evidence	Strength
Prototype → product-ready code	Postdoc: research RNN → Elk Audio OS VST on Raspberry Pi 4, 14 ms, 74× faster than DTW (JAES 2026). MAS: C++/Python real-time inference engines on embedded/IoT hardware	Strong
Optimize for platforms under performance/power/size limits	31.4% CPU on RPi4 within a 30 ms budget; compute and power profiling at McGill; Nebula C++17 library with per-feature ARM latency benchmarks	Strong (ARM/CPU). Weak on GPU/DSP
2+ CV areas from the list	**OCR**: tech-pack digitization (OCR + layout segmentation); multilingual care-label verification; COVID vitals reading digits off monitors. **Detection/recognition**: care-label template + feature matching, fabric defect detection, garment segmentation. Bonus **geometry/registration**: Promptly camera-to-print alignment; overlapping-FOV microscopic camera array	Strong (2 areas, a 3rd partial)
Strong C/C++	C++17 (Nebula), real-time engines, PLC/RIP integration for Aeoon DTG printer	Strong
Reusable algorithms for large-scale data	Nebula (reusable library); SSIM matcher repurposed from CV; 4 Zenodo datasets (9,839 files)	Medium
Applied maths	S-transform onset detection (Asilomar), SSIM, DTW, PhD-level benchmarking	Strong
Python prototyping	All published ML work	Strong

JD asks for	Your evidence	Strength
Code reviews, automated tests, architecture docs	Introduced CI/CD and code review at MAS; pytest; deployment docs for MUSMET partners	Strong
Work across embedded, backend, CV engineers	MAS: PLC, RIP and conveyor integration; MUSMET: 12-partner consortium, BCI + XR + audio sync with g.tec and SAE Barcelona	Strong
GenAI dev tools	Daily AI-assisted workflow; gave a talk on it	Medium
CNNs / ViTs / VLMs	CNN on symbolic matrices (FRUCT 2020); Wav2Vec2 transformer benchmarking (speakfrench); no ViT/VLM work	Partial
Parallel/vectorized compute, CUDA/OpenCL	Multi-GPU training (RTX 4090 + 3090) for VarianceEngine; no hand-written kernels or SIMD intrinsics on record	Gap
French	~A2, classroom study ongoing	Gap

Gaps and how to handle them

A recruiter screens for risk, not depth. Expect the first three; name each plainly and bridge in one sentence.

Gap	What the recruiter may worry about	Your line
Overqualified for Professional II	Flight risk; salary mismatch; boredom	"I'm applying for the work, not the title. Turning vision prototypes into fast C++ on cameras is what I want to be doing. If the team thinks my scope fits a higher level, I'm open to that conversation, but it isn't a condition."

Gap	What the recruiter may worry about	Your line
CV work is 2015–2020; recent work is audio	Rusty on vision	"The problem shape didn't change: real-time detection on constrained hardware. My audio models run on spectrograms, which are images, and one of my published methods repurposes SSIM from computer vision. Coming back to vision is a short step."
Relocation, 3 days/week onsite	Will you actually move?	"Yes. I'm in Toronto now, I'd relocate to Montreal, and three days in office is fine." Then ask about relocation support (section 7)
French ~A2	Quebec workplace language	"My French is basic and I'm actively studying it in classes. Day-to-day technical work in English is fine for me; I'd keep improving." Never say bilingual
No CUDA/OpenCL kernels, no SIMD intrinsics on record	Can't do the optimization half	"My optimization work has been CPU and ARM: profiling, memory and latency budgets on Raspberry Pi. I've trained on multi-GPU rigs but haven't written kernels. That's the area I'd ramp on first."
No SLAM / multi-view geometry	Missing a listed area	Don't volunteer. If asked: the list needs 2 of 8; you have OCR and detection/recognition, plus image registration from Promptly

Gap	What the recruiter may worry about	Your line
Gap since Jan 2026	Unemployed 9 months	"Postdoc ended December. Since then: relocation, two papers through review (Smart Drums, JAES in press; MIRaaS, IEEE IS² 2026), and independent projects — Nebula, a C++17 audio-feature library, and a French pronunciation scorer benchmarked over 100,000 trials."

The Jobalytics bar showed a 34% resume match. If the CV you submitted was an audio or generic variant, offer to send a CV-focused version after the call.

Pitch

Lead with MAS, not the PhD: for this recruiter, industrial vision in production is the headline and the audio work is proof you can still ship on hardware.

30 seconds (answer to "tell me about yourself")

> I'm an ML engineer with a PhD from the University of Trento and about ten years across industry and research. The thread through all of it is taking detection algorithms from prototype to code that runs in real time on constrained hardware. Before my PhD I spent five and a half years at MAS Holdings, South Asia's largest apparel manufacturer, building production computer-vision systems: an OpenCV care-label inspection line that cut inspection time by 99.5%, loom-side fabric defect detection, and the image-alignment algorithm behind an on-demand garment printing product now running in four countries. In my PhD and postdoc I did the same for audio, ending with a model that runs in 14 milliseconds on a Raspberry Pi 4, 74 times faster than the baseline. This role is that job for vision, on Zebra's cameras.

90 seconds (add these if they want more)

- **The MAS arc:** the care-label system went through three versions — flatbed scanner, industrial camera, then a conveyor with PLC reject. That progression is prototype-to-product.
- **Optimization habit:** every system I've shipped had a hard latency or compute budget. On the Pi it was 30 ms and about a third of the CPU.

- **Why Zebra Montreal:** the team behind the Aurora Imaging Library has been doing industrial vision since the Matrox days, which is the domain I came from. I also have ties to Montreal through my visiting research at McGill.
- **Engineering practice:** I set up CI/CD and code review at MAS and mentored engineers, so code review and automated tests are normal working habits for me.

Likely questions and answers

Keep each answer under a minute. The salary answer is the one that matters most on this call.

1. **Salary expectations?** The band is posted, so skip the "you go first" dance and anchor in its upper part: "Given the posted range and my experience, I'm targeting around $130K base, plus the bonus. I'm flexible on the total package, including relocation support." If pushed lower, ask where they would place you in the band and why, rather than naming a second number. Know your floor before the call.
2. **Why Zebra / why this role?** "Industrial machine vision is where I started, and I want to get back to shipping it. Your team owns the vision software stack and ships it onto real cameras — that's the prototype-to-product work I enjoy most."
3. **You're more senior than this level — why apply?** Use the overqualified line from the Gaps table. Then ask whether there's room on level (section 7).
4. **Are you open to Montreal, 3 days in office?** "Yes. I'd relocate from Toronto; three days onsite works."
5. **French level?** "Basic, around A2. I'm taking classes and keep improving. I can work in English day to day."
6. **Work authorization?** "Canadian permanent resident. No sponsorship needed."
7. **Availability / notice?** "Available immediately."
8. **Other processes?** "I'm at interview stage with a few other companies." Don't name them.
9. **Which CV areas from the list?** "OCR and object detection/recognition from production work at MAS, plus image registration for the garment-printing alignment."
10. **C++ level?** "Modern C++ — my most recent library is C++17, and I've written real-time engines and hardware integrations in C++."
11. **GPU / CUDA?** Use the honest line from the Gaps table. Don't stretch.
12. **Why leave research?** "My research was always engineering-heavy — everything I published ran on hardware. I want my work to ship to customers, not stop at a demo."
13. **What have you done since your postdoc?** Use the gap line from the Gaps table.

14. **Tell me about a time you turned a prototype into production.** Care-label QC: scanner → industrial camera → conveyor with PLC reject, inspection time from 15–20 min per label to 1–2 min per batch.

Questions to ask

Pick four or five; the first three are the ones that change your decision.

1. Which team is this — the Aurora Imaging Library group, smart cameras, or 3D? How many engineers are on it?
2. What's driving this hire, and how has the team changed over the past year? (Polite route to the 2026 restructuring.)
3. Is there flexibility on level? With a PhD and 10 years, could the scope fit a Professional III or Senior role?
4. Which hardware targets matter most right now — x86, NVIDIA GPU, ARM, or on-camera accelerators?
5. How is time split between algorithm work and optimization/productization?
6. What does the interview process look like after this call? Is there a C++ coding test or take-home, and who's on the panel?
7. What's the timeline to an offer?
8. In practice, what's the team's working language, and how much French is needed day to day?
9. Is relocation support offered for a move from Toronto?
10. Is the office the Dorval site, and are in-office days fixed or flexible?

Day-of checklist and red lines

- [ ] Open the JD, this guide and the exact CV you submitted
- [ ] Check which phone number that CV shows. Older variants carry the retired +1 817 number; give the recruiter +1 613 200 2030
- [ ] Check the recruiter's email is @zebra.com
- [ ] Know the product names: Aurora Imaging Library (ex-Matrox Imaging Library), Aurora Vision Studio, Aurora Design Assistant, Photoneo 3D
- [ ] Decide your salary floor before dialing
- [ ] Quiet room, headset, water
- [ ] Within 24 h: thank-you email; offer a CV-focused CV if the one on file is audio-heavy

Red lines

- Never cite ">90% F1". Use F1 0.76 (audio), 0.78 (gesture fusion) or 95% detection (SSIM).
- Never mention the Promptly patent. Your role: alignment algorithm, RIP integration, mechanism consulting.
- Don't claim CUDA/OpenCL kernels, SIMD intrinsics or ViT/VLM experience.
- Don't describe your French as conversational or bilingual.
- No SIN, banking or ID details by phone or email.

Sources

- Zebra Q2 2026 results — 8-K Exhibit 99.1
- Zebra 10-Q, quarter ended 4 Jul 2026 (restructuring, robotics sale)
- Zebra 10-Q, quarter ended 28 Jun 2025 (Photoneo acquisition)
- Q4 2025 earnings call transcript — The Motley Fool
- Matrox Imaging acquisition — Wiley Industry News
- Aurora deep-learning tools — BusinessWire
- WARN filing, San Jose, Feb 2026 — WARNTracker
- Anonymous layoff thread, Feb 2026 — TheLayoff (unverified)
- Zebra employee reviews — Glassdoor